

JUDE RUSSELL

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EDUCATION

Imperial College London <i>MSc in Physics with Quantum Dynamics</i>	Sept 2025 – Sept 2026
King's College London <i>BSc in Physics with a Year Abroad, First Class Honours</i>	Sept 2021 – May 2025
University of California, Berkeley <i>Physics exchange year</i>	Aug 2023 – May 2024

RESEARCH EXPERIENCE

Quantum Measurement Lab, Imperial College London <i>Master's Research Project – Dr Michael Vanner</i> - Measured the multimode Brillouin spectrum of standard optical fibre using heterodyne spectroscopy, resolving five closely spaced acoustic resonances. - Developed analytical and numerical models of acoustic confinement to investigate the observed mode spacing and predict its temperature dependence. - Predicted how neighbouring acoustic-mode spacings respond to temperature, showing that much of the common Brillouin frequency shift cancels in their difference.	Apr 2026 – Sept 2026
Atto@Kings, King's College London <i>Undergraduate Research Project – Dr Emilio Pisanty and Dr Anne Weber</i> - Extended a numerical simulation framework to investigate electron trajectories and quantum-path phase coefficients in two-colour high-harmonic generation. - Calculated intensity-dependent phase responses and investigated possible signatures of quantum tunnelling without a classical barrier. - Identified distinctive phase-sensitive features in bichromatic fields, providing possible directions for subsequent experimental investigation.	Jan 2025 – Apr 2025
Levitated Nanophysics Group, King's College London <i>King's Undergraduate Research Fellowship – Professor James Millen</i> - Investigated the gradual loss of trapping stability of microscopic diamond particles in a modified Paul trap. - Developed MATLAB particle-tracking software and used power spectral analysis to characterise the evolution of particle motion. - Used simplified numerical modelling to investigate whether a reduction in charge-to-mass ratio could explain the observed frequency shift.	Jun 2023 – Aug 2023

SELECTED PROJECTS

Laser spectroscopy for magneto-optical trapping <i>Experimental project, Imperial College London</i> Aligned free-space optics and recorded Doppler-free saturated-absorption spectra of the rubidium D₂ line. Investigated laser-frequency drift in a system used for cooling-laser stabilisation.	Jul 2026
Surface-GKP Quantum Error Correction <i>Technical review, Imperial College London</i> Reviewed GKP encoding, syndrome extraction, decoding, finite squeezing and the concatenation of bosonic GKP codes with surface codes, including possible experimental implementations.	Jan 2026 – Mar 2026

Ground-state energy estimation with SQD

Oct 2025 – Nov 2025

IBM Qiskit Fall Fest Team Project

- Applied sample-based quantum diagonalisation to estimate the electronic ground-state energy of a water molecule.
- Compared the resulting estimates with a classical reference and presented the project at an Institute of Physics event.

Cal Aerospace SAE, UC Berkeley

Sept 2023 – May 2024

Structures Team

- Contributed to the structural design and analysis of a 15 ft competition aircraft, using SolidWorks and Ansys finite-element analysis to investigate landing-gear performance under heavy landing loads.

In2scienceUK

Jun 2020 – Aug 2020

Summer Intern

- Investigated the optical properties of lead-halide perovskites and analysed spectral data to identify band-gap trends.

PRESENTATIONS

*Oral presentations***Quantum computing for drug discovery**

Nov 2025

*Institute of Physics, London***Quantum tunnelling without a barrier: phase-sensitive signatures in HHG**

May 2025

*King's College London***Dancing levitated diamonds in a modified Paul trap**

Aug 2023

*Levitated nanophysics group, King's College London**Poster presentations***Predicting the temperature dependence of multimode Brillouin scattering in optical fibre**

Aug 2026

*Blackett Laboratory Foyer, Imperial College London***Quantum tunnelling without a barrier?**

May 2025

*Undergraduate poster conference, King's College London***TECHNICAL SKILLS**

Programming: Python, Julia, MATLAB, Mathematica, LaTeX.**Experimental:** Optical alignment, laser spectroscopy, heterodyne detection, photodetection, electrical spectrum analysis and high-voltage electronics.**Numerical methods:** Eigenvalue calculations, finite-element modelling, Fourier analysis, optimisation, particle tracking and scientific data analysis.**Quantum software:** Qiskit, QuTiP, IBM Quantum Platform.**AWARDS & FUNDING**

King's Research Award

Dec 2023

*King's College London – undergraduate research***King's Undergraduate Research Fellowship**

May 2023

*Competitive funded summer research programme***Turing Fund**

May 2023

*Funding for study abroad in the United States***Quantum Mechanics Module Prize**

Dec 2022

King's College London – top performance in a cohort of 120